

Table 8: Prompt used to evaluate whether a paper is relevant to the topic

You are an expert researcher tasked with evaluating whether a given paper is relevant to our research topic based on its title and abstract.

Below are the details of the paper you need to assess:

Title: **[Title]**

Abstract: **[Abstract]**

The topic is: **[Topic]**

If the paper title and abstract are related to the topic, output 1; otherwise, output 0. As long as you feel that this article has reference value for your question, you can use it to help you study the topic, it does not need to be completely consistent in topic.

Please follow the strict format below:

Think: ...

Relevant: 0/1

Table 9: Prompt used to extract idea, experiment, entities and references from paper (part I)

You are a scientific research expert, tasked with extracting and summarizing information from provided paper content relevant to the topic: **[Topic]**. Your deliverables will include pertinent references, extracted entities, a detailed summary, and the experimental design.

The topic you are studying is: **[Topic]** (Ensure that the references are pertinent to this topic.)

Extraction Requirements:

Entities:

1. Identify unique entities mentioned in the paper, such as model names, datasets, metrics, and specialized terminology.
2. Format the entities with a name followed by a brief description.
3. Ensure all entities are relevant to the specified topic (**[Topic]**).

Summary Idea:

1. Background: Elaborate on the task's context and previous work, outlining the starting point of this paper.
2. Novelty: Describe the main innovations and contributions of this paper in comparison to prior work.
3. Contribution: Explain the primary methods used, detailing the theory and functions of each core component.
4. Detail Reason: Provide a thorough explanation of why the chosen methods are effective, including implementation details for further research.
5. Limitation: Discuss current shortcomings of the approach.

Continue to next table →

Table 10: Prompt used to extract idea, experiment, entities and references from paper (part II)

Experimental Content:

1. Experimental Process: Detail the entire experimental procedure, from dataset construction to specific steps, ensuring clarity and thoroughness.
2. Technical Details: Describe any specific technologies involved, providing detailed implementation processes.
3. Clarity of Plan: State your experimental plan concisely to facilitate understanding without unnecessary complexity.
4. Baseline: Elaborate on the baseline used, comparative methods, and experimental design, illustrating how these support and validate the conclusions drawn.
5. Verification: Explain how your experimental design assists in verifying the core idea and ensure it is detailed and feasible.

Relevance Criteria:

1. Method Relevance: References must directly correlate with the paper’s methodology, indicating improvements or modifications.
2. Task Relevance: References should address the same task, even if methods differ, better have the same topic **[Topic]**
3. Baseline Relevance: References should serve as baselines for the methods discussed in the paper.
4. Output Format: Provide references without author names or publication years, formatted as titles only.

The paper content is as follows: **[Paper content]**

Please provide the entities, summary idea, experimental design, and the three most relevant references (Sort by relevance, with priority given to new ones with the same level of relevance, do not reference the original paper.) based on the paper’s content.

Note: Ensure the references are pertinent to the topic you are studying: **[Topic]**. If there are no relevant references, output [].

Now please output strictly in the following format:

Entities: ...

Idea: ...

Experiment: ...

References: ...

Table 11: Prompt used to get trends of CoI

You are a scientific research expert tasked with summarizing the historical progression of research related to our current topic, based on the literature we have reviewed.

Here are the entities you need to know : **[Entities]**
 The topic you are studying is: : **[Topic]**
 The literature from early to late: **[Idea chain]**
 Your objective is to outline the historical evolution of the research in light of current trends. Please follow these requirements:

Analysis of Published Viewpoints: Examine the progression of ideas across the identified papers. Detail how each paper transitions to the next--for instance, how Paper 0 leads to Paper 1, and so forth. Focus on understanding how Paper 1 builds upon the concepts in Paper 0. Elaborate on specific advancements made, including proposed modules, their designs, and the rationale behind their effectiveness in addressing previous challenges. Apply this analytical approach to each paper in the sequence.

Please present your findings in the following format:

Trends:
 Paper 0 to Paper 1: ...
 Paper 1 to Paper 2: ...
 ...

Table 12: Prompt used to predict future trend (Part I)

You are a scientific expert tasked with formulating a novel and innovative research idea based on your comprehensive literature review. Your objective is to propose a feasible approach that could significantly advance the field.

Here are the entities you need to know : **[Entities]**
 The literature you have studied is as follows: **[Chain of ideas]**
 The following section delineates the progressive relationships among the previously summarized research papers: **[Trend]**
 Based on previous research, analyze how human experts think and transition from previous methods to subsequent approaches. Focus on their reasoning logic and the sources of their thought processes. Learn to emulate their reasoning patterns to further develop and guide your own research direction in a natural and coherent manner.

Additionally, you are encouraged to adopt the following three modes of thinking:

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